

Machine Learning: The No-Nonsense (But Fun) Guide — Day 10

Decision Tree Regression + Pre-Pruning + Mini Project 🌳📊

1. Story & Intuition: Predicting House Prices

Yesterday we classified flowers. Today, we **predict numbers**—like house prices.

Imagine you're a real estate agent. A client asks:

"How much is this house worth?"

You look at:

- Square footage
- Number of bedrooms
- Location score

And you think:

"Hmm, houses in this neighborhood with 3 beds usually go for around \$400k."

That's **Decision Tree Regression**. Instead of predicting categories, it predicts **continuous values** by averaging the numbers in each leaf.

2. Decision Tree Regression: How It Works

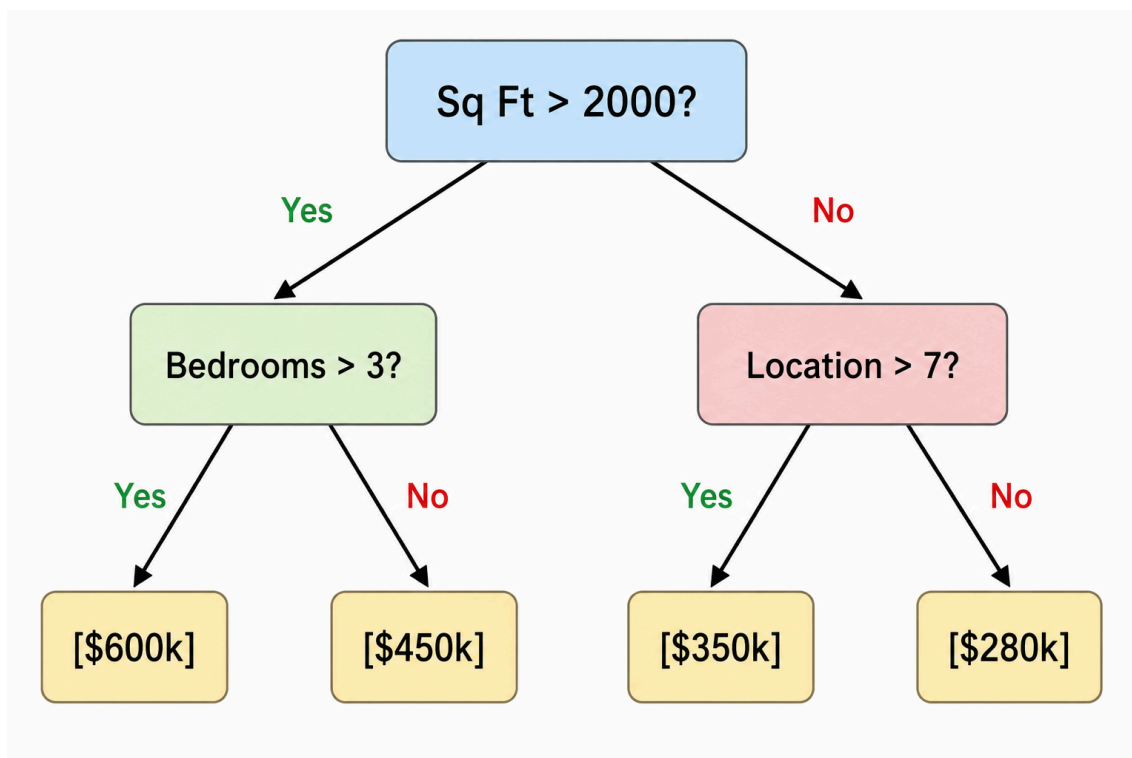
Classification Tree

Leaf = Majority Vote → **"Cat"** or **"Dog"**

Regression Tree

Leaf = Average Value → **"\$450,000"**

Regression Tree for House Prices



↑ Average price

↑ Average price

Splitting Criteria

Metric	Formula	Goal
Mean Squared Error (MSE)	Average of (Actual – Predicted) ²	Minimize variance

Mean Absolute Error (MAE)	Average of Actual – Predicted	Reduce effect of outliers
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How the Tree Chooses a Split

- Try every possible split.
 - Calculate the MSE after each split.
 - Select the split with the **lowest MSE**.
 - Continue until a stopping condition is reached.
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3. Pre-Pruning

Why Do We Need It?

Problem: Trees continue growing until they perfectly memorize the training data, causing **overfitting**.

Solution: Stop the tree early by setting limits before training.

Hyperparameters

Hyperparameter	Purpose	Recommended Range
max_depth	Maximum depth	3–10
min_samples_split	Minimum samples before splitting	5–20
min_samples_leaf	Minimum samples in each leaf	2–10
max_leaf_nodes	Maximum number of leaves	10–50

min_impurity_decrease	Minimum improvement needed	0.01
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4. Pre-Pruning vs Post-Pruning

Feature	Pre-Pruning	Post-Pruning
When	Before growth	After growth
Speed	Faster	Slower
Control	Easy	More precise
Risk	Stops early	May initially overfit

Best Practice: Begin with **Pre-Pruning** and fine-tune later using **Post-Pruning** if necessary.

5. Visualizing the Tree

- Each node shows the splitting feature.
 - **Samples** = Number of observations.
 - **Value** = Average target value.
 - **MSE** = Error within that node.
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6. Mini Project – Predicting Car Prices 🚗

Step 1 – Create Dataset

Step 2 – Data Preprocessing

Step 3 – Train Models

Results

=== Overfit Tree ===

...

=== Pre-Pruned Tree ===

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Observation: The pre-pruned tree performs better on unseen data.

Try to do a mini project from scratch.

You can check my mini project from github for reference !!

7. Interview Questions

1. Difference between Decision Tree Classifier and Regressor?
 2. Which metric is used for Regression Trees?
 3. What is Pre-Pruning?
 4. Why use `min_samples_leaf`?
 5. Why does Pre-Pruning reduce overfitting?
 6. Difference between Pre-Pruning and Post-Pruning?
 7. What is feature importance?
 8. Can Decision Trees handle categorical and numerical data?
 9. What happens if `max_depth` is too large or too small?
 10. How does GridSearchCV optimize a Decision Tree?
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Happy Learning !!!